Manufacturing Defect Prediction

Using Logistic Regression on Unified Production Data

Intern: Dhara Nayak

Data Science Internship 2026

01 | Business Problem & Objective

BUSINESS PROBLEM

Manufacturing plants face significant losses due to undetected product defects that reach downstream processes or customers.

- No early warning system for high scrap batches
- Defects detected too late - after production is complete
- Manual inspection is inconsistent and costly
- Root cause across 6 data systems is unknown

PROJECT OBJECTIVE

Build a machine learning model that predicts whether a production batch will be defective, enabling proactive quality intervention.

- Consolidate data from 6 operational systems
- Identify key drivers of scrap & rework
- Classify each batch: Defect (1) or Normal (0)
- Achieve high Recall - minimize missed defects
- Deliver actionable insights to reduce scrap cost

02 | Data Overview & Consolidation

ERP

Planned units, shifts, production schedule

MES

Good units, downtime, changeover, overtime

QMS

Scrap %, rework minutes, quality records

SCADA

Speed RPM, pressure, vibration, oil temp

Machine

Machine type, criticality, vendor, sensor

Operator

Skill level, experience, training hours

Data preparation flow :- *Merge 6 sources → Standardise columns → Remove duplicates → Verify completeness → Unified Database*

UNIFIED PRODUCTION DATASET
12,203 rows × 38 columns

03 | Feature Preparation & Analysis

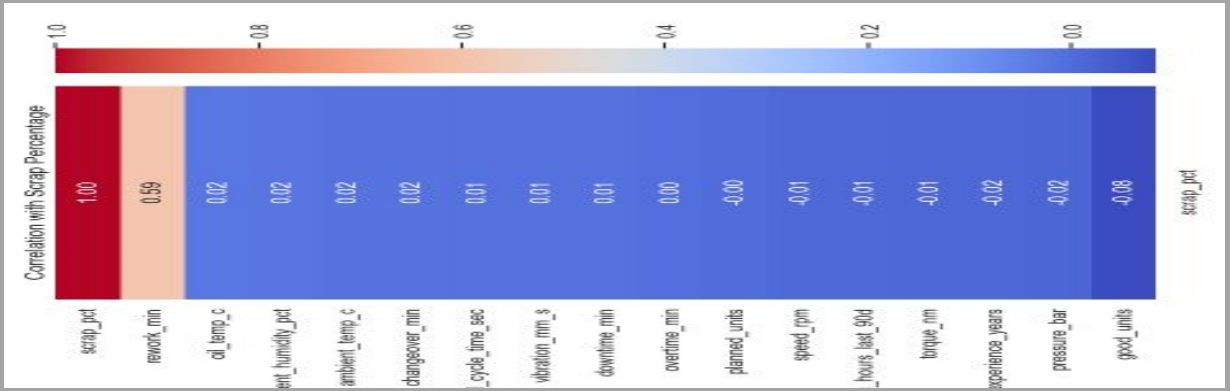
MISSING VALUE TREATMENT

- Handled missing data to ensure model stability:
- Numerical : Filled with median (robust to outliers)
- Categorical : Filled with most frequent value
- Final dataset had no missing values

OUTLIER DETECTION & TREATMENT

- Outliers were capped (not removed) to avoid data loss:
- Used IQR method to detect extreme values
- Applied winsorization to limit extreme impact
- Key affected features: rework_min, experience_years
-

CORRELATION WITH TARGET (scrap_pct)



KEY INSIGHT

rework_min is the only HIGH-correlation feature (r = 0.573). More rework → more scrap.

04 | Results & Validation

Confusion Matrix:

Improved Logistic Regression | Test Set: 2,441 rows

	Predicted: Normal	Predicted: Defect
Actual: Normal	1,175 True Negative	50 False Positive
Actual: Defect	107 False Negative	1,109 True Positive

Model Performance Metrics:

Accuracy : 93.6%
Precision : 95.7%
Recall : 91.2%
F1 Score : 93.4%
ROC-AUC : 98.8%

Sample: Actual vs Predicted:

Row	Actual	Predicted	Result
1	Normal	Normal	✓
2	Normal	Normal	✓
3	Defect	Defect	✓
5	Normal	Defect	✗

RECOMMENDATION

Model is production-ready. Deploy to flag high-risk batches proactively. Focus process improvement on rework_min — the #1 defect driver.

05 | Final Recommendation

MODEL VERDICT

1. Deploy Model
Use model in MES/ERP to flag high-risk batches early

2. Reduce Rework
Focus on rework_min to lower defect rates

3. Monitor Model
Track performance and retrain regularly

Thank You